

# ACHINTYA RAI

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## EDUCATION

### University of California, San Diego

Bachelor of Science in Computer Engineering

- NSF REU Fellow, UC Scholars Research Fellow, Competitive Programming Club

Sept. 2024 – May 2027

GPA: 3.9/4.0

## WORK EXPERIENCE

### Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Designed a time-series anomaly detector by iterating statistical automation algorithms to select the optimal machine learning model, handling seasonality across **10+** workflows and **30+** countries.
- Developed SQL and MLOps data pipelines to schedule weekly retraining and hourly inference, scaling the anomaly detector across all payment forecasting models.
- Profiled **60.6M+** payment events in Databricks, surfacing **86,387** duplicate records and a **47%** null rate across **1,272** monitoring segments to benchmark the machine learning detector.

### Decklar (formerly Roambee)

Software Engineering Intern

Sunnyvale, CA

June 2024 – Aug. 2024

- Optimized Python data pipelines for real-time shipment monitoring at Amazon, Coca-Cola, and Samsung, boosting capacity by **15%** for **1M+** weekly events.
- Implemented visualization tools for analytics modules, accelerating reporting speeds by **20%** and improving data integrity across shipment dashboards.

## RESEARCH

### Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Engineered a Python machine learning pipeline to extract LLM embedding manifolds (Llama-2, DeepSeek-V2), compressing **500GB** model checkpoints to **13GB** localized datasets.
- Optimized preprocessing scripts to achieve a **30x** runtime reduction, enabling large-scale Deep Learning analysis on resource-constrained **8GB RAM** hardware.

### Data Science Teaching & Learning Lab

Undergraduate Researcher

San Diego, CA

Sept. 2024 – Present

- Developed a generative AI labeling workflow using the Gemini API to automate classification of **16k+** student data points, reducing labeling time by **90%**.
- Architected a multi-agent artificial intelligence tutoring framework in Python with hallucination-reduction strategies, achieving **95%+** accuracy across **100k+** data points.

## PROJECTS

### NanoGPT | Python, PyTorch, C++, Transformers, LLM Optimization | [🔗](#)

Mar. 2026

- Implemented a GPT-style transformer in Python and PyTorch with Deep Learning techniques (multi-headed attention, backpropagation), sustaining **100%** GPU utilization during training.
- Compiled critical sub-modules in C++ and optimized memory layout, leveraging GPU technology to reduce per-token latency by **25%**.

### SquaredUpChess | React, Django, AWS, Docker, WebSockets, Coolify, SQLite | [🔗](#)

Aug. 2025 – Dec. 2025

- Deployed full-stack platform on AWS using Docker, architecting **20+** REST endpoints with a CI/CD pipeline for **99.9%** deployment reliability.

## TECHNICAL SKILLS

**Languages:** Python, C++, TypeScript, SQL (Postgres/SQLite), Java, JavaScript, Bash, HTML/CSS

**Frameworks:** PyTorch, React, Django, FastAPI, Docker, Flask, Asyncio, WebSockets

**AI & Cloud:** Deep Learning, Machine Learning, AWS, GCP, GPU technology, Gemini/OpenAI API, Linux